

Chapter 11

Linear Approximations and Differentials

§ Section: 3.10

11.1 Linear Approximations

$$y = f(x)$$

$$y = f(a) + f'(a) \cdot (x - a)$$

We use the tangent line at $(a, f(a))$ as an approximation to the curve $y = f(x)$ when x is near a .

Definition:

$f(x) \approx f(a) + f'(a)(x - a)$ is called the **linear approximation** or **tangent line approximation** of f at a . And we also define $L(x) = f(a) + f'(a)(x - a)$, which is called the **linearization** of f at a .

Example 1.1

Write down the linearization of $f(x) = x^{\frac{1}{5}}$ at $a = 1$. Estimate $(0.999)^{\frac{1}{5}}$ by the linear approximation.

Sol: The linearization of $f(x)$ at $a=1$ is

$$L(x) = f(1) + f'(1)(x-1), \text{ where}$$

$$f(1) = 1^{\frac{1}{5}} = 1, \quad f'(1) = \frac{1}{5} \cdot 1^{-\frac{4}{5}} = \frac{1}{5}.$$

$$\Rightarrow L(x) = 1 + \frac{1}{5}(x-1)$$

$$(0.999)^{\frac{1}{5}} = f(0.999) \approx L(0.999) = 0.9998$$

Example 1.2

To use linearizations to estimate the value $f(b)$, we need to find a number a that is close to b and compute the linearization of $f(x)$ at a , $L(x)$. Then we can approximate $f(b)$ by $L(b)$.

$f(b)$	$\sin(0.002)$	$\sqrt{82}$	$\ln(1.02)$
$f(x)$	$\sin x$	$\sqrt{x} = x^{\frac{1}{2}}$	$\ln x$
Choose $a =$. Linearization of $f(x)$ at a	Choose $a = 0$ $\underline{L(x) = f(0) + f'(0)(x - 0)}$ $= 0 + (x - 0) = \underline{x}$ $\sin x \approx x$ when x is near 0 .	Choose $a = 81$ $L(x) = f(81) + f'(81)(x - 81)$ $= 9 + \frac{1}{18}(x - 81)$	Choose $a = 1$ $L(x) = f(1) + f'(1)(x - 1)$ $= x - 1.$
Approximate $f(b)$ by $L(b)$	$\sin(0.002) \approx L(0.002)$ $= 0.002$	$\sqrt{82} = f(82) \approx L(82)$ $= 9 + \frac{1}{18} = 9.0\bar{5}$	$\ln(1.02) \approx L(1.02)$ $= 0.02$
Investigate the error $f(b) - L(b)$.	$\sin 0.002 - 0.002$ $\approx -1.3 \times 10^{-9}$	$\sqrt{82} - 9.0\bar{5}$ $= -0.00017 \dots$	$\ln(1.02) - 0.02$ $\approx -0.0002 \dots$

Remark:

The accurate value of previous questions is

- $(0.999)^{\frac{1}{5}} = 0.9997999199519664 \dots$
- $\sin(0.002) = 0.0019999986666669333 \dots$
- $\sqrt{82} = 9.055385138137417 \dots$
- $\ln(1.02) = 0.01980262729618 \dots$

Example 1.3 $\sin x \cdot y^3 + (x+1)y = 1$

$\sin x \cdot y^3 + xy = 1$ defines y implicitly as a function of x near $(0, 1)$, say $y = f(x)$. Use the linear approximation to estimate $f(-0.05)$. $f(0) = 1$

Sol: Compute $f'(0) = \frac{dy}{dx} \Big|_{x=0}$.

$$\sin x \cdot (y(x))^3 + (x+1)y(x) = 1.$$

$$\Rightarrow \frac{d}{dx} \cos x \cdot y^3 + \sin x \cdot 3 \cdot y^2 \cdot y' + y + (x+1) \cdot y' = 0$$

$$\text{At } (0, 1), \quad 1 + 1 + y' = 0 \Rightarrow y' = -2$$

The linearization of $f(x)$ at $a=0$ is

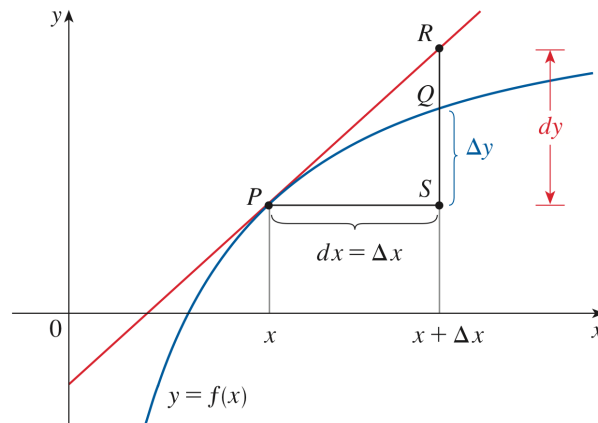
$$L(x) = f(0) + f'(0) \cdot (x-0) = 1 - 2x$$

$$f(-0.05) \approx L(-0.05) = 1.1$$

11.2 Differentials

If $y = f(x)$, where f is a differentiable function, then the differential dx is an independent variable; that is, dx can be any real number. The differential dy is then defined in terms of dx and x by the equation $dy = f'(x)dx$. So dy is a dependent variable; it depends on the values of x and dx . We can use dy to estimate the change of f due to the change of the variable x by an amount dx .

$$\Delta y \approx dy$$



dy represents the amount that the tangent line rises or falls (the change in the linearization), whereas Δy represents the amount that the curve $y = f(x)$ rises or falls when x changes by an amount dx .

Example 2.1

The radius of a sphere is measured and found to be 20 cm with possible error at most 0.05 cm. Approximate the maximum error of the volume.

sol:

$$V(r) = \frac{4}{3}\pi r^3$$

when $r = 20$, $dr = \Delta r = \pm 0.05$,

$$\begin{aligned} \text{estimate } \Delta V &\approx dV = V'(r) \cdot dr \Big|_{\substack{r=20 \\ dr=\pm 0.05}} \\ &= 4\pi r^2 \cdot dr \\ &= 4\pi \times 20^2 \times (\pm 0.05) = \pm 80\pi \end{aligned}$$

$$\text{relative error } \frac{\Delta V}{V} \approx \frac{dV}{V} = \pm \frac{80\pi}{\frac{4}{3}\pi(20)^3} = \pm \frac{3}{400}$$

How good is the linear approximation?

It seems that linear approximations are easy and efficient approximations! However, we need to further estimate the error $R_1(x) = f(x) - L(x) = f(x) - [f(a) + f'(a)(x - a)]$. We can show

that $\lim_{x \rightarrow a} \frac{f(x) - L(x)}{x - a} = 0$. (Handwritten: $\varepsilon(x)$ above the limit, and $\frac{f(x) - f(a) - f'(a)(x-a)}{x-a} = \lim_{x \rightarrow a} \left(\frac{f(x) - f(a)}{x-a} - f'(a) \right) = f'(a) - f'(a) = 0$)

$f(x) = L(x) + R_1(x) = L(x) + \varepsilon(x) \cdot (x-a)$, where $\varepsilon(x) \rightarrow 0$ as $x \rightarrow a$.

In fact, with the help of Cauchy's Mean Value Theorem, we can express the error in terms of $f''(x)$ and $|x - a|$:

$$R_1(x) = f(x) - L(x) = \frac{f''(c)}{2}(x - a)^2$$

for some c between x and a .

Example 2.2

Prove the chain rule $\left. \frac{d}{dx} f(g(x)) \right|_{x=a} = f'(g(a)) \cdot g'(a)$.